Lab 2: The Value of Prediction

Signals, Errors, and When to Follow the Machine

ECON 470 — Microeconomics of Artificial Intelligence

Spring 2026

1 Overview

In Lectures 3 and 4 you learned that AI is economically valuable because it resolves uncertainty — but only when it changes what the decision-maker does. In this lab you will work through the 2×2 prediction model from Gans, Chapter 3, applying it to a medical diagnosis setting and a spam filter setting. You will compute posteriors, enumerate strategies, and see exactly when and why following a signal is optimal.

Groups: 2–3 students. **Only one submission per group, please.**

Submit: A writeup with your answers to the numbered questions below. Show your work for all calculations.

Points: 100 total. See point values next to each question.

Time estimate: 3–4 hours. Start early — the graphing problems take longer than you expect.

If you get stuck: The Prediction Tutor can help you work through problems without giving away the answer: https://chatgpt.com/g/g-69d815bc1cdc81919933d59a1f8dd044-value-of-ai-prediction-tutor. Use it sparingly — you'll learn more by wrestling with the questions yourself first.

2 Problem 1: Medical Diagnosis and the Value of Prediction

Return to the medical diagnosis example from Lecture 3. A physician sees a patient presenting with symptoms of a respiratory infection. Based on the clinical presentation, the physician believes the infection is present ($\theta = 1$) with probability $p = 0.7$. The alternative is that the patient does not have the infection ($\theta = 2$).

The physician must choose an action: treat ($a = 1$) or not treat ($a = 2$). If the action matches the state, the payoff is $R = 100$; if it does not match, the payoff is $r = 20$.

An AI diagnostic tool produces a signal $s \in \{1, 2\}$ with accuracy $e = 0.85$. That is, $\Pr(s = 1 \mid \theta = 1) = 0.85$ and $\Pr(s = 2 \mid \theta = 2) = 0.85$.

2.1 Part A: Is the signal informative?

(1) Using the joint probability table (Table 3.2 in Gans), write out the four joint probabilities $\Pr(\theta = i, s = j)$ for this example. Verify they sum to 1. *[2 pts]*

(2) Compute $\Pr(\theta = 1 \mid s = 1)$, the probability that the patient is truly infected given a positive test result. Compare it to the prior $p = 0.7$. Does receiving $s = 1$ make the physician *more* confident the infection is present? *[3 pts]*

(3) Compute $\Pr(\theta = 1 \mid s = 2)$, the probability that the patient is truly infected given a negative test result. Compare it to $p = 0.7$. Does receiving $s = 2$ make the physician *more* confident the infection is not present? *[3 pts]*

(4) In one or two sentences, explain what it means to say the signal is "informative." Connect your answer to the two posteriors you just computed. *[2 pts]*

2.2 Part B: Expected payoff without a signal

(5) Without the signal, the physician follows a rule: always choose the action that matches the more likely state. What action is that? Compute the expected payoff under this rule and the expected payoff under the rule according to which the physician chooses the action that matches the least likely state. *[3 pts]*

2.3 Part C: Strategies with a signal

With a signal, the physician observes s before choosing a. A **strategy** is a complete plan: for each possible signal, which action to take.

(6) Enumerate all four possible strategies. For each one, describe in words what the physician does. *[3 pts]*

(7) For **each** of the four strategies, compute the expected payoff. To do this, compute the payoff in each of the four (θ, s) cells, weight by the joint probability from Part A, and sum. Organize your work in a table. *[8 pts]*

(8) Which strategy yields the highest expected payoff? Explain how this result is consistent with the condition $e \geq p$. *[3 pts]*

2.4 Part D: The value of the signal

(9) Compute the value of the AI diagnostic tool: the difference between the expected payoff under the best strategy from Part C and the expected payoff under the (best) rule from Part B. *[3 pts]*

3 Problem 2: The Demand for AI Diagnostic Services

Suppose there are 1,000 physicians, each facing the same type of diagnosis problem as in Problem 1: $p = 0.7$, $e = 0.85$. The physicians differ in the stakes of their decisions. For physician i, the cost of a misdiagnosis is $R_i - r_i$, which is distributed uniformly on the interval $[10, 200]$ across the population. (The physician from Problem 1, with $R - r = 80$, is one of them.)

An AI diagnostic service is available at price P per use.

3.1 Part A: Willingness to pay

(10) What is the willingness to pay (WTP) for the AI service as a function of $R_i - r_i$? *[3 pts]*

(11) What is the WTP of the physician with the lowest stakes ($R_i - r_i = 10$)? The highest stakes ($R_i - r_i = 200$)? The physician from Problem 1 ($R_i - r_i = 80$)? *[2 pts]*

3.2 Part B: Demand curve

(12) A physician adopts the AI service if her WTP $\geq P$. Derive the **cutoff** value of $R_i - r_i$ above which a physician adopts, as a function of P. *[4 pts]*

(13) Using the uniform distribution on $[10, 200]$, derive $D(P)$, the number of physicians (out of 1,000) who adopt the AI service at price P. For what range of P is demand strictly between 0 and 1,000? *[5 pts]*

(14) Plot $D(P)$ with price P on the horizontal axis and quantity (number of adopting physicians) on the vertical axis. Label the intercepts. *[3 pts]*

3.3 Part C: Better AI

(15) Now suppose a more accurate AI becomes available, with $e = 0.90$ (instead of 0.85). Derive the new demand curve $D'(P)$ and plot it **on the same graph** as $D(P)$, in a different color. *[3 pts]*

(16) At a price of $P = 12$, how many physicians adopt the old AI ($e = 0.85$)? How many adopt the new AI ($e = 0.90$)? *[2 pts]*

(17) In one or two sentences, explain the economic logic of the shift. Why does better prediction increase demand at every price? Connect your answer to the formula for the value of information. *[3 pts]*

4 Problem 3: Biased Signals and the Spam Filter

Return to the spam filter example from Lecture 3. An email system must classify messages as not-spam ($a = 1$) or spam ($a = 2$). The probability a message is legitimate is $p = 0.95$. Payoffs are $R = 1$ (correct classification) and $r = 0.5$ (incorrect classification).

Now suppose the signal can have **different accuracy in each state**:

- $e_1 = \Pr(s = 1 \mid \theta = 1)$ — sensitivity (how often the test correctly identifies legitimate email)
- $e_2 = \Pr(s = 2 \mid \theta = 2)$ — specificity (how often the test correctly identifies spam)

The available prediction technology has average accuracy $\bar{\lambda} = 0.70$, so the agent faces the constraint $\frac{e_1+e_2}{2} = 0.70$. Each e_i must also lie in $[0, 1]$.

The agent follows the signal when the agent chooses $a = 1$ for $s = 1$ and $a = 2$ for $s = 2$.

4.1 Part A: Expected payoff as a function of e_1 and e_2

(18) Write out the expected payoff of following the signal as a function of e_1 and e_2, using the joint probabilities from Table 3.3 in Gans with $p = 0.95$, $R = 1$, and $r = 0.5$. Simplify. *[5 pts]*

4.2 Part B: The accuracy frontier

(19) Draw the constraint $\frac{e_1+e_2}{2} = 0.70$ in a graph with e_2 on the horizontal axis and e_1 on the vertical axis. Remember that each e_i must lie in $[0, 1]$, so the feasible portion of the line is capped at 1 on each axis. Label the endpoints. *[3 pts]*

(20) On the same graph, sketch several indifference curves of the expected payoff function from Part A. Indicate the direction in which expected payoff increases. *[4 pts]*

(21) Using your graph, identify the optimal choice of (e_1^*, e_2^*). Mark it on the graph. What are the numerical values? *[3 pts]*

4.3 Part C: Interpretation

(22) Does this agent prioritize sensitivity over specificity, or is it the other way around? Fully explain in simple terms why this would be so. What does this mean in terms of the spam filter's behavior? *[4 pts]*

(23) Connect your answer to the "relative to default" intuition from lecture. Without any signal, the agent's default action is $a = 1$ (mark everything as not-spam). What is the incremental cost of a false positive (spam gets through) relative to this default? What is the incremental cost of a false negative (legitimate email flagged as spam)? *[4 pts]*

4.4 Part D: The ROC curve

(24) Redraw your graph, this time with the **false positive rate** (FPR $= 1 - e_2$) on the horizontal axis and the **true positive rate** (TPR $= e_1$) on the vertical axis. Plot the feasible set (the set of achievable (FPR, TPR) pairs given the constraint), the suitably redrawn indifference map, and mark the optimal point. *[4 pts]*

(25) Compare your graph to Figure 3.1 in Gans. What role does the accuracy frontier play in the ROC diagram? What does improving the prediction technology (increasing $\bar{\lambda}$) do to the feasible set in this diagram? *[3 pts]*

5 Problem 4: Reading the ROC Curve

The figure below shows three ROC curves: one for random prediction (red), one for an OK prediction technology (blue), and one for perfect prediction (green).

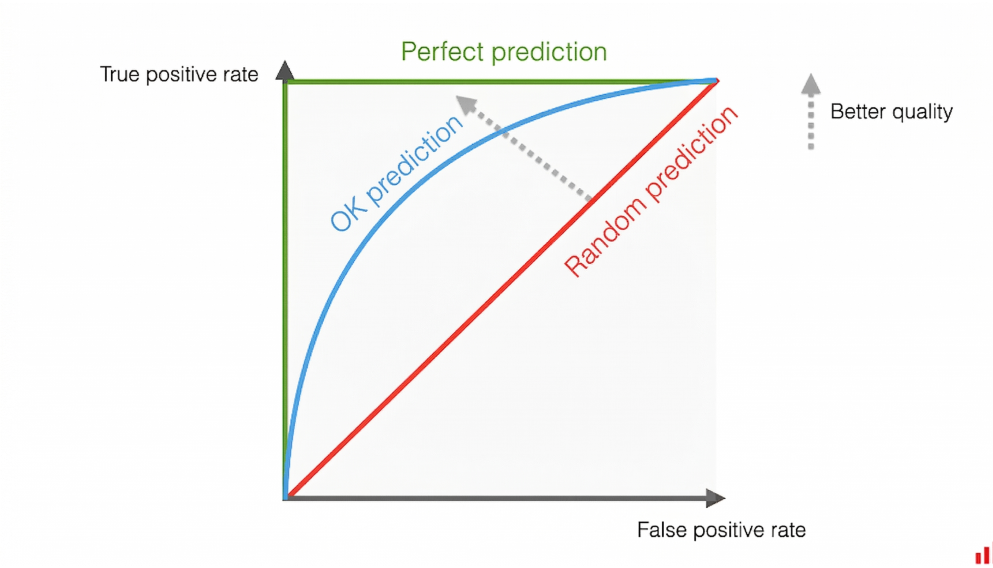

Figure 1: ROC curves for three prediction technologies.

(26) **The red line (random prediction).** The 45-degree line represents a signal with $\bar{\lambda} = 0.50$, i.e., $e_1 + e_2 = 1$. Show algebraically that this implies TPR = FPR for every point on this line. In plain language, explain why a signal where TPR = FPR is useless: what does it mean about the signal's ability to distinguish between the two states? *[4 pts]*

(27) **The green line (perfect prediction).** The green L-shape runs vertically from $(0, 0)$ to $(0, 1)$ and then horizontally from $(0, 1)$ to $(1, 1)$. This corresponds to $\bar{\lambda} = 1$, i.e., $e_1 + e_2 = 2$. Explain why the upper-left corner $(0, 1)$ — where FPR = 0 and TPR = 1 — represents a perfect signal. What are e_1 and e_2 at that point? *[4 pts]*

(28) **The blue curve (OK prediction).** The blue curve lies between the red and green lines. As the agent moves along this curve from the lower-left to the upper-right, what is being traded off? Why must an agent with an imperfect prediction technology ($0.5 < \bar{\lambda} < 1$) accept this trade-off? *[4 pts]*

(29) **Improving prediction.** Using the language of this diagram, explain what it means for AI to "get better" over time. What happens to the ROC curve? *[3 pts]*